Worked Answer Key

Page 1

1. $48 \div 6 = 8$ stickers each. Check: $6 \times 8 = 48$.
2. $7 \times 8 = 56$ chairs; related division fact: $56 \div 7 = 8$ (or $56 \div 8 = 7$).

Page 2

1. $63 \div 9 = 7$ because $9 \times 7 = 63$.
2. $96 \div 12 = 8$ because $12 \times 8 = 96$.

Page 3

1. $735 \div 5 = 147$; $5 \times 147 = 735$.
2. $864 \div 8 = 108$; $8 \times 108 = 864$.

Page 4

1. Area model: split an area of 924 into $880 + 44 = (22 \times 40) + (22 \times 2)$. With one side 22, the other side is $40 + 2 = 42$, so $924 \div 22 = 42$.
2. Area model: split an area of 1,176 into $1,120 + 56 = (28 \times 40) + (28 \times 2)$. With one side 28, the other side is $40 + 2 = 42$, so $1,176 \div 28 = 42$.

Page 5

1. $4,032 \div 6 = 672$; $6 \times 672 = 4,032$.
2. $5,040 \div 7 = 720$; $7 \times 720 = 5,040$.

Page 6

1. $157 \div 6 = 26$ R 1; $6 \times 26 + 1 = 157$, and $1 < 6$.
2. $389 \div 9 = 43$ R 2; $9 \times 43 + 2 = 389$, and $2 < 9$.

Page 7

1. $2,365 \div 14 = 168$ R 13; check $14 \times 168 + 13 = 2,352 + 13 = 2,365$.
2. $4,118 \div 25 = 164$ R 18; check $25 \times 164 + 18 = 4,100 + 18 = 4,118$.

Page 8

1. $74 \div 8 = 9$ R 2. Nine vans hold 72; two students need a tenth van, so **10 vans**.
2. $74 \div 8 = 9$ R 2: **9 full bags and 2 apples left**.

Page 9

1. $1,248 \div 24 = 52$ eggs per carton; $24 \times 52 = 1,248$.
2. $936 \div 18 = 52$ boxes; $18 \times 52 = 936$.

Page 10

1. $527 \div 32 = 16$ R 15: **16 full shelves, 15 books remain**.
2. $1,005 \div 40 = 25$ R 5: **25 full bundles, 5 flyers remain**.

Page 11

1. Estimate $2,916 \div 7 \approx 2,800 \div 7 = 400$. Exact: $2,916 \div 7 = 416$ R 4.
2. Estimate $6,284 \div 31 \approx 6,200 \div 31 = 200$. Exact: $6,284 \div 31 = 202$ R 22.

Page 12

1. **392**; $3,528$ is close to $3,600$, and $3,600 \div 9 = 400$. Thus 392 is reasonable, while 3,920 is much too large. Check: $9 \times 392 = 3,528$.
2. **300** is the reasonable estimate: $8,190$ is close to $8,100$, and $8,100 \div 27 = 300$. In contrast, $27 \times 30 = 810$, far below $8,190$.

Page 13

1. Digit sum $= 4 + 5 + 7 + 2 = 18$. Divisible by 3 and 9; even, so also divisible by 6.
2. Ends in 5, so divisible by 5. Does not end in 0, so not divisible by 10. Last two digits 25, so divisible by 25.

Page 14

1. $18 \times 47 = 846$, so the missing number is **846**.
2. $2,736 \div 76 = 36$, so the missing divisor is **36**; $36 \times 76 = 2,736$.

Page 15

1. $1,440 \div 12 = 120$ and $1,440 \div 15 = 96$; $120 > 96$.
2. Both equal 150: $2,400 \div 16 = 150$ and $3,000 \div 20 = 150$; use $=$.

Page 16

1. $15 \times 28 = 420$ seats; $420 \div 7 = 60$ seats per section.
2. $9 \times 36 = 324$ muffins; $324 \div 12 = 27$ muffins per box.

Page 17

1. $\$84 \div 6 = \$14$ per person.
2. $3,456 \div 24 = 144$ meters per section.

Page 18

1. Jordan is correct: $8 \times 85 + 5 = 680 + 5 = 685$. The answer is **85 R 5**; the remainder 5 is less than 8.
2. Incorrect. Estimate: $1,248 \div 16 \approx 1,280 \div 16 = 80$, so 708 is far too large. The correct quotient is **78**, since $16 \times 78 = 1,248$.

Page 19

1. $7,245 \div 15 = 483$; check $15 \times 483 = 7,245$.
2. $5,678 \div 24 = 236$ R 14; check $24 \times 236 + 14 = 5,664 + 14 = 5,678$.

Page 20

1. $2,375 \div 48 = 49$ R 23. Forty-nine full buses carry 2,352 campers; **50 buses** are needed, with **23 campers on the last bus**.
2. $1,234 \times 9 + 7 = 11,106 + 7 = 11,113$. The number **11,113** has five digits; $11,113 \div 9 = 1,234$ R 7, and $7 < 9$ makes the remainder valid.